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Cloud RAN (C-RAN): A Technology Overview

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Introduction

Mobile data traffic has been growing at an explosive rate, driven largely by the proliferation of smartphones and tablets. As spectral efficiency in existing standards such as LTE approaches theoretical limits, operators are forced to densify networks by deploying more cells, which sharply increases both capital expenditure (CAPEX) and operating expenditure (OPEX). At the same time, average revenue per user (ARPU) remains flat, creating a widening gap between network cost and operator revenue. Cloud Radio Access Network (C-RAN) has emerged as a promising architecture to address this imbalance by centralizing and virtualizing baseband processing resources.

What is C-RAN?

C-RAN evolves from the traditional base station architecture, in which radio and baseband functions are co-located, and from the later Remote Radio Head (RRH) architecture, which separates the radio unit from the baseband unit (BBU) and connects them over fiber. C-RAN takes this separation further by pooling the BBUs from many cell sites into a centralized, virtualized "BBU Pool" or "BBU Hotel," while RRHs remain at the antenna sites. The RRH-to-BBU link, known as the fronthaul, typically uses the Common Public Radio Interface (CPRI) to transport in-phase and quadrature (IQ) samples under strict bandwidth, latency, and jitter constraints.

Key Advantages

Centralizing baseband processing yields a statistical multiplexing gain: because traffic peaks in different cells (e.g., office versus residential areas) occur at different times of day, a shared pool requires less total processing capacity than the sum of individually dimensioned base stations, with studies cited in the paper reporting savings in the range of roughly 20–75% in BBU count or compute resources depending on assumptions and traffic mix. This translates into substantial energy and cost savings, with the paper citing estimates of around 15% CAPEX and up to 50% OPEX reduction relative to traditional RAN. Co-location of BBUs also facilitates advanced LTE-Advanced coordination features such as enhanced Inter-Cell Interference Coordination (eICIC) and Coordinated Multi-Point (CoMP) transmission/reception, both of which rely on tight synchronization that is easier to achieve when processing resources sit in one physical pool rather than being distributed across many sites. Additional benefits include faster, lower-delay handovers performed within the pool rather than across the X2 interface between separate eNodeBs, and simplified network maintenance and upgrades, since hardware interventions are concentrated in a small number of centralized locations rather than at every cell site.

Challenges

Realizing C-RAN in practice is constrained by three main challenges. First, the fronthaul network must support very high bandwidth alongside strict latency and jitter budgets — for example, sub-frame processing delay must stay below 1 ms to satisfy HARQ timing, and round-trip user-data delay is generally capped at a few microseconds. Second, effective techniques for BBU cooperation, interconnection, and clustering are still maturing, since coordination features like CoMP require sharing user data and channel feedback across cells. Third, virtualizing the BBU Pool as a coherent, elastic computing cluster — analogous to a data-center cloud — remains an active area of research and standardization.

Conclusion

C-RAN offers a compelling path toward more energy-efficient, cost-effective, and adaptable mobile networks by decoupling radio and baseband functions and pooling processing resources in a virtualized, centralized cloud. While it is not the only architecture competing to meet growing capacity demands — small cells and Massive MIMO remain relevant alternatives, particularly where fiber is scarce — C-RAN is especially attractive to operators with access to cheap fiber infrastructure and is widely regarded as a foundational building block for future 5G soft, green network architectures.

Cloud RAN for Mobile Networks—A Technology Overview

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Abstract—Cloud Radio Access Network (C-RAN) is a novel mobile network architecture which can address a number of challenges the operators face while trying to support growing end-user's needs. The main idea behind C-RAN is to pool the Baseband Units (BBUs) from multiple base stations into centralized BBU Pool for statistical multiplexing gain, while shifting the burden to the high-speed wireline transmission of In-phase and Quadrature (IQ) data. C-RAN enables energy efficient network operation and possible cost savings on baseband resources. Furthermore, it improves network capacity by performing load balancing and cooperative processing of signals originating from several base stations. This paper surveys the state-of-the-art literature on C-RAN. It can serve as a starting point for anyone willing to understand C-RAN architecture and advance the research on C-RAN.

Index Terms—Cloud RAN, mobile networks, small cells, eICIC, CoMP, virtualization, in-phase and quadrature (IQ) compression, CPRI.

I. INTRODUCTION

MOBILE data transmission volume is continuously rising. It is forecasted to grow 13-fold from 2012 until 2017 according to Cisco [1], with smart phones and tablet users driving the growth. Therefore, to satisfy growing user demands, mobile network operators have to increase network capacity. As spectral efficiency for the Long Term Evolution (LTE) standard is approaching the Shannon limit, the most prominent way to increase network capacity is by either adding more cells, creating a complex structure of Heterogeneous and Small cell Networks (HetSNets) [2] or by implementing techniques such as multi-user Multiple Input Multiple Output (MIMO) [3] as well as Massive MIMO [4], where numerous antennas simultaneously serve a number of users in the same time-frequency resource. However, this results in growing inter-cell interference levels and high costs.

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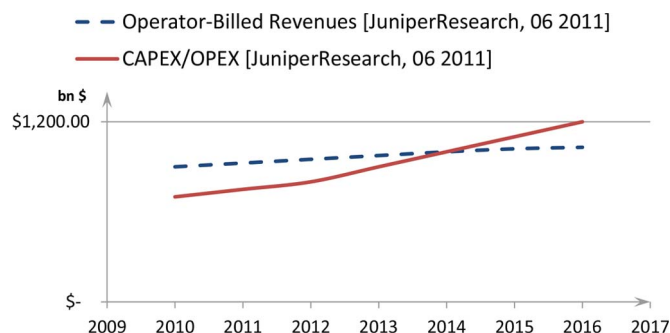


Fig. 1. Costs versus revenues in mobile networks.

Total Cost of Ownership (TCO) in mobile networks includes CAPITAL EXpenditure (CAPEX) and OPERating EXpenditure (OPEX). CAPEX mainly refers to expenditure relevant to network construction which may span from network planning to site acquisition, RF hardware, baseband hardware, software licenses, leased line connections, installation, civil cost and site support, like power and cooling. OPEX covers the cost needed to operate the network, i.e., site rental, leased line, electricity, operation and maintenance as well as upgrade [5]. CAPEX and OPEX are increasing significantly when more base stations are deployed. More specifically, CAPEX increases as base stations are the most expensive components of a wireless network infrastructure, while OPEX increases as cell sites demand a considerable amount of power to operate, e.g., China Mobile estimates 72% of total power consumption originates from the cell sites [6]. Mobile network operators need to cover the expenses for network construction, operation, maintenance and upgrade; meanwhile, the Average Revenue Per User (ARPU) stays flat or even decreases over time, as the typical user is more and more data-hungry but expects to pay less for data usage. As presented in Fig. 1 [7], mobile operators are facing cases (2014–2015) where network cost may exceed revenues if no remedial actions are taken [8]. Therefore, novel architectures that optimize cost and energy consumption become a necessity in the field of mobile network.

Cloud radio access network (C-RAN) is a novel mobile network architecture, which has the potential to answer the above mentioned challenges. The concept was first proposed in [9] and described in detail in [6]. In C-RAN, baseband processing is centralized and shared among sites in a virtualized BBU Pool. This means that it is able to adapt to non-uniform traffic and utilizes the resources, i.e., base stations, more efficiently. Due to that fact that fewer BBUs are needed in C-RAN compared to the traditional architecture, C-RAN has

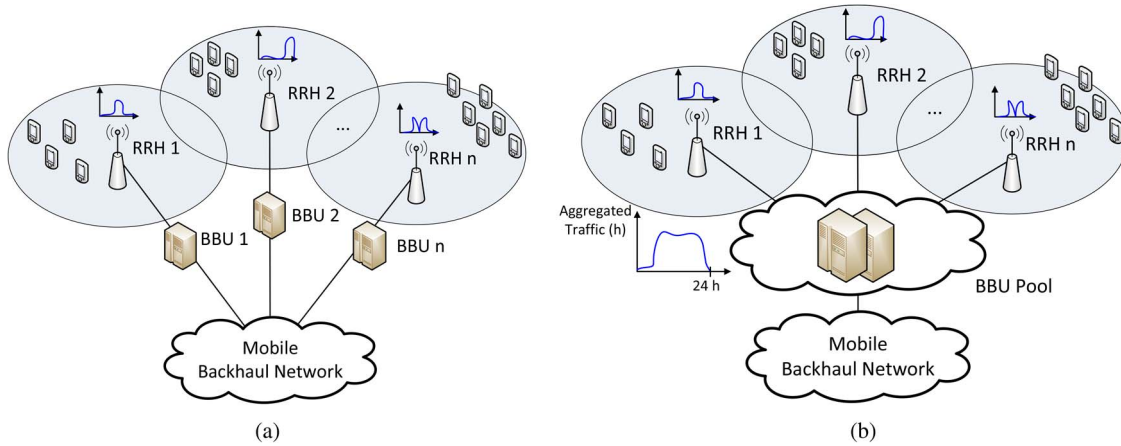


Fig. 2. Statistical multiplexing gain in C-RAN architecture for mobile networks. (a) RAN with RRH. (b) C-RAN.

also the potential to decrease the cost of network operation, because power and energy consumption are reduced compared to the traditional RAN architecture. New BBU can be added and upgraded easily, thereby improving scalability and easing network maintenance. Virtualized BBU Pool can be shared by different network operators, allowing them to rent Radio Access Network (RAN) as a cloud service. As BBU from many sites are co-located in one pool, they can interact with lower delays—therefore mechanisms introduced for LTE-Advanced (LTE-A) to increase spectral efficiency and throughput, such as enhanced ICIC and Coordinated Multi-Point (CoMP) are greatly facilitated. Methods for implementing load balancing between the cells are also facilitated. Furthermore, network performance is improved, e.g., by reducing delay during intra-BBU Pool handover.

C-RAN architecture is targeted by mobile network operators, as envisioned by China Mobile Research Institute [6], IBM [9], Alcatel-Lucent [10], Huawei [11], ZTE [12], Nokia Siemens Networks [5], Intel [13] and Texas Instruments [14]. Moreover, C-RAN is seen as typical realization of mobile network supporting soft and green technologies in fifth generation (5G) mobile network in year 2020 horizon [15]. However, C-RAN is not the only candidate architecture that can answer the challenges faced by mobile network operators. Other solutions include small cells, being part of HetSNets and Massive MIMO. Small cells deployments are the main competitors for outdoor hot spot as well as indoor coverage scenarios. All-in-one small footprint solutions like Alcatel-Lucent's LightRadio can host all base station functionalities in a few liters box. They can be placed outdoors reducing cost of operation associated to cooling and cell site rental. However, they will be underutilized during low-activity periods and can not employ collaborative functionalities as well as C-RAN can do. Moreover, they are more difficult to upgrade and repair than C-RAN. Brief comparison between C-RAN, Massive MIMO and HetSNets is outlined in [2]. In [16], Liu *et al.* proves that energy efficiency of large scale Small Cell Networks is higher compared with Massive MIMO. Furthermore, cost evaluation on different options needs to be performed in order for a mobile network operator to choose an optimal solution. Comparison of TCO including CAPEX and OPEX over 8 years of traditional LTE macro base station, LTE C-RAN and LTE small cell shows that the total transport

cost per Mb/s is highest for macro cell deployment—2200\$, medium for C-RAN—1800\$ and 3 times smaller for small cell—600\$ [17]. Therefore the author concludes that C-RAN needs to achieve significant benefits to overcome such a high transportation cost. Collaborative techniques such as CoMP and eICIC can be implemented in small cells giving higher benefits in HetNet configuration instead of C-RAN. The author envisions that C-RAN might be considered for special cases like stadium coverage. However, C-RAN is attractive for operators that have free/cheap fiber resources available.

This article surveys the state-of-the-art literature published on C-RAN and its implementation. Such input helps mobile network operators to make an optimal choice on deployment strategies. The paper is organized as follows. In Section II we introduce the fundamental aspects of C-RAN architecture. Moreover, in Section III we discuss in detail the advantages of this architecture along with the challenges that need to be overcome before fully exploiting its benefits in Section IV. In Section V we also present a number of constraints in regards to the transport network capacity imposed by C-RAN and discuss possible solutions, such as the utilization of compression schemes. In Sections VI, VII we give an overview of the state-of-the-art hardware solutions that are needed to deliver C-RAN from the radio, baseband and network sides. As the BBU Pool needs to be treated as a single entity, in Section VIII we present an overview of virtualization techniques that can be deployed inside a BBU Pool. In Section IX we evaluate possible deployment scenarios of C-RAN. In Section X we summarize ongoing work on C-RAN and give examples of first field trials and prototypes. Section XI concludes the paper.

II. WHAT IS C-RAN? BASE STATION ARCHITECTURE EVOLUTION

C-RAN is a network architecture where baseband resources are pooled, so that they can be shared between base stations. Fig. 2 gives an overview of the overall C-RAN architecture. This section gives an introduction to base station evolution and the basis of the C-RAN concept.

The area which a mobile network covers is divided into cells, therefore mobile networks are often called cellular networks. Traditionally, in cellular networks, users communicate with

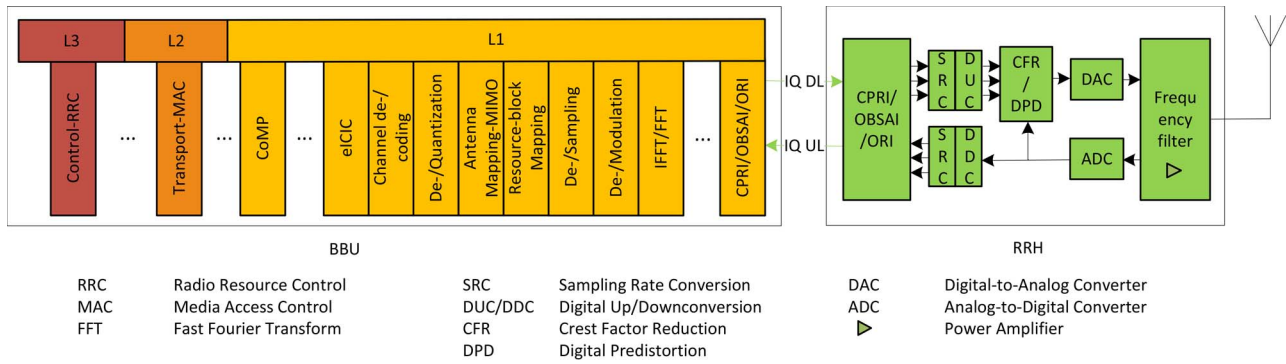


Fig. 3. Base station functionalities. Exemplary baseband processing functionalities inside BBU are presented for LTE implementation. Connection to RF part and sub modules of RRH are shown.

a base station that serves the cell under coverage of which they are located. The main functions of a base station can be divided into baseband processing and radio functionalities. The main sub-functions of baseband processing module are shown in left side of Fig. 3. Among those we find coding, modulation, Fast Fourier Transform (FFT), etc. The radio module is responsible for digital processing, frequency filtering and power amplification.

A. Traditional Architecture

In the traditional architecture, radio and baseband processing functionality is integrated inside a base station. The antenna module is generally located in the proximity (few meters) of the radio module as shown in Fig. 4(a) as coaxial cables employed to connect them exhibit high losses. X2 interface is defined between base stations, S1 interface connects a base station with mobile core network. This architecture was popular for 1G and 2G mobile networks deployment.

B. Base Station With RRH

In a base station with Remote Radio Head (RRH) architecture, the base station is separated into a radio unit and a signal processing unit, as shown in Fig. 4(b). The radio unit is called a RRH or Remote Radio Unit (RRU). RRH provides the interface to the fiber and performs digital processing, digital to analog conversion, analog to digital conversion, power amplification and filtering [18]. The baseband signal processing part is called a BBU or Data Unit (DU). More about BBU can be found in Chapter 16 of [19]. Interconnection and function split between BBU and RRH are depicted in Fig. 3. This architecture was introduced when 3G networks were being deployed and right now the majority of base stations use it.

The distance between a RRH and a BBU can be extended up to 40 km, where the limitation is coming from processing and propagation delay. Optical fiber and microwave connections can be used. In this architecture, the BBU equipment can be placed in a more convenient, easily accessible place, enabling cost savings on site rental and maintenance compared to the traditional RAN architecture, where a BBU needs to be placed close to the antenna. RRHs can be placed up on poles or rooftops, leveraging efficient cooling and saving on air-conditioning in BBU housing. RRHs are statically assigned to BBUs similarly to the traditional RAN. One BBU can serve

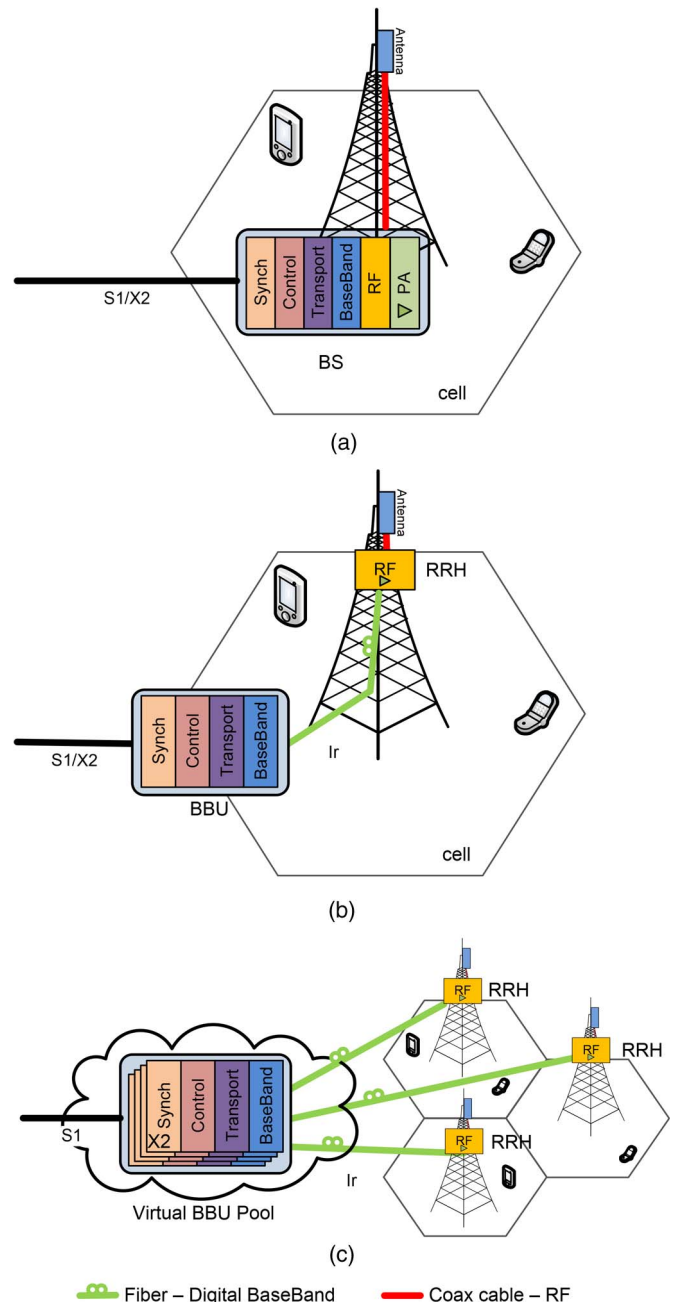


Fig. 4. Base station architecture evolution. (a) Traditional macro base station. (b) Base station with RRH. (c) C-RAN with RRHs.

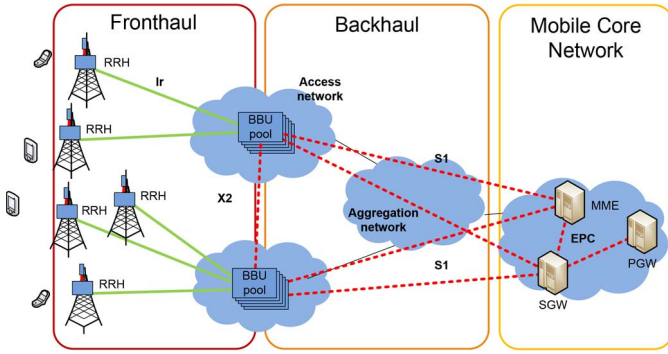


Fig. 5. C-RAN LTE mobile network.

many RRHs. RRHs can be connected to each other in a so called daisy chained architecture. An Ir interface is defined, which connects RRH and BBU.

Common Public Radio Interface (CPRI) [20] is the radio interface protocol widely used for IQ data transmission between RRHs and BBUs—on Ir interface. It is a constant bit rate, bidirectional protocol that requires accurate synchronization and strict latency control. Other protocols that can be used are Open Base Station Architecture Initiative (OBSAI) [21] and Open Radio equipment Interface (ORI) [22], [23].

C. Centralized Base Station Architecture—C-RAN

In C-RAN, to optimize BBU utilization between heavily and lightly loaded base stations, the BBUs are centralized into one entity that is called a BBU/DU Pool/Hotel. A BBU Pool is shared between cell sites and virtualized as shown in Fig. 4(c). A BBU Pool is a virtualized cluster which can consist of general purpose processors to perform baseband (PHY/MAC) processing. X2 interface in a new form, often referred to as X2+ organizes inter-cluster communication.

The concept of C-RAN was first introduced by IBM [9] under the name Wireless Network Cloud (WNC) and builds on the concept of Distributed Wireless Communication System [24]. In [24] Zhou *et al.* proposes a mobile network architecture in which a user communicates with densely placed distributed antennas and the signal is processed by Distributed Processing Centers (DPCs). C-RAN is the term used now to describe this architecture, where the letter C can be interpreted as: Cloud, Centralized processing, Cooperative radio, Collaborative or Clean.

Fig. 5 shows an example of a C-RAN mobile LTE network. The fronthaul part of the network spans from the RRHs sites to the BBU Pool. The backhaul connects the BBU Pool with the mobile core network. At a remote site, RRHs are co-located with the antennas. RRHs are connected to the high performance processors in the BBU Pool through low latency, high bandwidth optical transport links. Digital baseband, i.e., IQ samples, are sent between a RRH and a BBU.

Table I compares traditional base station, base station with RRH and base station in C-RAN architecture.

III. ADVANTAGES OF C-RAN

Both macro and small cell can benefit from C-RAN architecture. For macro base station deployments, a centralized BBU Pool enables an efficient utilization of BBUs and reduces

TABLE I
COMPARISON BETWEEN TRADITIONAL BASE STATION,
BASE STATION WITH RRH AND C-RAN

Architecture	Radio and baseband functionalities	Problem it addresses	Problems it causes
Traditional base station	Co-located in one unit	-	High power consumption Resources are underutilized
Base station with RRH	Spitted between RRH and BBU. RRH is placed together with antenna at the remote site. BBU located within 20-40 km away. Generally deployed nowadays	Lower power consumption. More convenient placement of BBU	Resources are underutilized
C-RAN	Spitted into RRH and BBU. RRH is placed together with antenna at the remote site. BBUs from many sites are co-located in the pool within 20-40 km away. Possibly deployed in the future	Even lower power consumption. Lower number of BBUs needed - cost reduction	Considerable transport resources between RRH and BBU

the cost of base stations deployment and operation. It also reduces power consumption and provides increased flexibility in network upgrades and adaptability to non-uniform traffic. Furthermore, advanced features of LTE-A, such as CoMP and interference mitigation, can be efficiently supported by C-RAN, which is essential especially for small cells deployments. Last but not least, having high computational processing power shared by many users placed closer to them, mobile operators can offer users more attractive Service Level Agreements (SLAs), as the response time of application servers is noticeably shorter if data is cached in BBU Pool [25]. Network operators can partner with third-party service developers to host servers for applications, locating them in the cloud—in the BBU Pool [26]. In this section we describe and motivate advantages of C-RAN: A. Adaptability to nonuniform traffic and scalability, B. Energy and cost savings, C. Increase of throughput, decrease of delays as well as D. Ease in network upgrades and maintenance.

A. Adaptability to Nonuniform Traffic and Scalability

Typically, during a day, users are moving between different areas, e.g., residential and office. Fig. 6 illustrates how the network load varies throughout the day. Base stations are often dimensioned for busy hours, which means that when users move from office to residential areas, the huge amount of processing power is wasted in the areas from which the users have moved. Peak traffic load can be even 10 times higher than during off-the-peak hours [6]. In each cell, daily traffic distribution varies, and the peaks of traffic occur at different hours. Since in C-RAN baseband processing of multiple cells is carried out in the centralized BBU pool, the overall utilization rate can be improved. The required baseband processing capacity of the pool is expected to be smaller than the sum of capacities of single base stations. The ratio of sum of single base stations capacity to the capacity required in the pool is called statistical multiplexing gain.

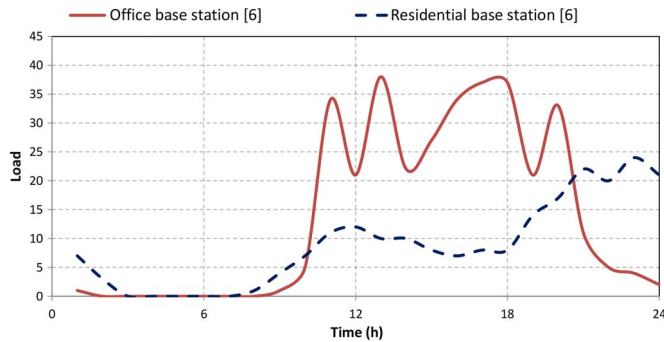


Fig. 6. Daily load on base stations varies depending on base station location.

In [27] an analysis on statistical multiplexing gain is performed as a function of cell layout. The analysis shows that in the Tokyo metropolitan area, the number of BBUs can be reduced by 75% compared to the traditional RAN architecture. In [28] Madhavan *et al.* quantifies the multiplexing gain of consolidating WiMAX base stations in different traffic conditions. The gain increases linearly with network size and it is higher when base stations are experiencing higher traffic intensity. In our previous work [29] we present initial evaluation of statistical multiplexing gain of BBUs in C-RAN. The paper concludes that 4 times less BBUs are needed for user data processing in a C-RAN compared to a traditional RAN for specific traffic patterns, making assumptions of the number of base stations serving different types of areas. The model does not include mobile standard protocols processing. After including protocol processing in [30] we concluded that the statistical multiplexing gain varies between 1.2 and 1.6 depending on traffic mix, thereby enabling saving of 17%–38%. In [31] Bhaumik *et al.* shows that the centralized architecture can potentially result in savings of at least 22% in compute resources by exploiting the variations in the processing load across base stations. Results have been evaluated experimentally. In [32] Werthmann *et al.* proves that the data traffic influences the variance of the compute resource utilization, which in consequence leads to significant multiplexing gains if multiple sectors are aggregated into one single cloud base station. Aggregation of 57 sectors in a single BBU Pool saves more than 25% of the compute resources. Moreover, the user distribution has a strong influence on the utilization of the compute resources. The results of last three works converge giving around 25% of potential savings on baseband resources.

Statistical multiplexing gain can be maximized by employing a flexible, reconfigurable mapping between RRH and BBU adjusting to different traffic profiles [33]. Statistical multiplexing gain depends on the traffic, therefore it can be maximized by connecting RRHs with particular traffic profiles to different BBU Pools [30].

Coverage upgrades simply require the connection of new RRHs to the already existing BBU Pool. To enhance network capacity, existing cells can then be split, or additional RRHs can be added to the BBU Pool, which increases network flexibility. Deployment of new cells is in general more easily accepted by local communities, as only a small device needs to be installed on site (RRH) and not a bulky base station. If the overall network capacity shall be increased, this can be easily achieved

by upgrading the BBU Pool, either by adding more hardware or exchanging existing BBUs with more powerful ones.

As BBUs from a large area will be co-located in the same BBU Pool, load balancing features can be enabled with advanced algorithms on both the BBU side and the cells side. On the BBU side, BBUs already form one entity, therefore load balancing is a matter of assigning proper BBU resources within a pool. On the cells side, users can be switched between cells without constraints if the BBU Pool has capacity to support them, as capacity can be assigned dynamically from the pool.

B. Energy and Cost Savings Coming From Statistical Multiplexing Gain in BBU Pool

By deploying C-RAN, energy, and as a consequence, cost savings, can be achieved [34]. Eighty percent of the CAPEX is spent on RAN [6], therefore, it is important to work towards reducing it.

Energy in mobile network is spent on power amplifiers, supplying RRH and BBU with power and air conditioning. Forty-one percent of OPEX on a cell site is spent on electricity [6]. Employing C-RAN offers potential reduction of electricity cost, as the number of BBUs in a C-RAN is reduced compared to a traditional RAN. Moreover, in the lower traffic period, e.g., during the night, some BBUs in the pool can be switched off not affecting overall network coverage. Another important factor is the decrease of cooling resources, which takes 46% of cell site power consumption [6]. Due to the usage of RRHs air conditioning of radio module can be decreased as RRHs are naturally cooled by air hanging on masts or building walls, as depicted in Fig. 4. ZTE estimates that C-RAN enables 67%–80% power savings compared with traditional RAN architecture, depending on how many cells one BBU Pool covers [12], which stays in line with China Mobile research claiming 71% power savings [35].

Civil work on remote sites can be reduced by gathering equipment in a central room, what contributes to additional OPEX savings.

In total, 15% CAPEX and 50% OPEX savings are envisioned comparing to RAN with RRH [35] or traditional RAN architecture [36]. However, the cost of leasing the fiber connection to the site may increase CAPEX. IQ signal transported between RRHs and BBUs brings up a significant overhead. Consequently, the installation and operation of transport network causes considerable costs for operators.

C. Increase of Throughput, Decrease of Delays

The next generation mobile network, envisaged to eventually replace the 3G networks is called LTE and has been standardized by Third Generation Partnership Project (3GPP) (in Release 8 and onwards of the standards). See [37] for a comprehensive overview. LTE-A is the latest mobile network standard prepared by the 3GPP in Release 10–12 of the standards. Any mobile network standard could potentially be deployed in a C-RAN architecture. However, as LTE is currently deployed all over the world, LTE and LTE-A are the most prominent standards to be deployed as C-RAN. This section introduces LTE radio access scheme and mechanisms proposed for

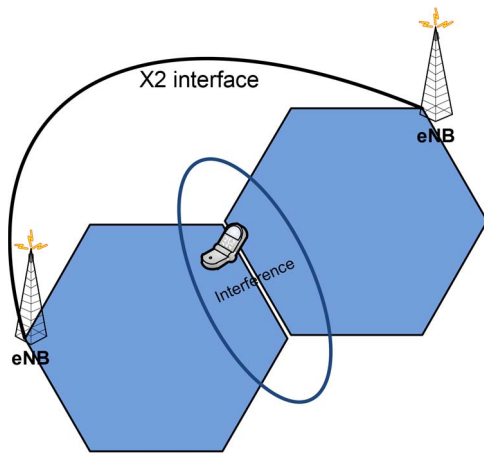


Fig. 7. Interference handling in LTE network.

LTE-A—eICIC and CoMP. Because of pooling of BBU resources in a C-RAN, those features are greatly facilitated, as signal processing from many cells can be done over one BBU Pool, easing the implementation and reducing processing and transmitting delays. Good understanding of eICIC and CoMP helps to conclude about the opportunities that C-RAN offers.

LTE operates with shared resources only. There is a scheduler in the base station (called evolved Node B (eNB) in LTE) that takes care of all resource allocation/assignments. A key feature in LTE is the radio access scheme based on Orthogonal Frequency-Division Multiple Access (OFDMA). The basic idea in OFDMA is to use a large number of densely spaced, orthogonal carriers. Resources can be dynamically allocated both in the frequency and time domain. This gives a very flexible utilization of the available resources.

LTE systems generally use a frequency reuse factor of 1, meaning that all cells operate at the same frequency. Hence, inter-cell interference is particularly high in such systems. This is observed as a very high ratio (up to a factor of 10) between peak throughput and cell edge throughput.

Basically, there are two approaches to address the interference issue: minimizing interference and exploiting interference paths constructively.

1) *Minimizing Inter Cell Interference*: Inter cell interference can be avoided either statically or dynamically in time, frequency and power domain. An obvious, static solution is not to use co-channel deployment, i.e., simply by using different frequencies in adjacent cells. This is called hard frequency reuse, and has the advantage that it avoids X2 signaling almost entirely. Fractional frequency reuse can also be used (static and dynamic approaches are commonly used, see e.g., [38]). However, as the frequency resources on lower bands are scarce it is better to use other solutions rather than the ones involving frequency reuse. Therefore, this section focuses on the case where the same frequency resources are being used in all cells.

In Release 8, Inter-cell Interference Coordination (ICIC) was introduced. In this scheme UEs can report back to the eNB in case they experience strong interference on certain sub-carriers. The eNB can then (by using the X2 interface) coordinate with the neighboring cell so that these sub-carriers are not used for that particular mobile, as shown in Fig. 7. It is important to note

here, that this is applied to cell-edge mobiles only. Near the center of the cell there is no interference and the full resource (i.e., entire frequency band) set can be utilized.

The scheme works in uplink (UL) as well as downlink (DL). In DL the eNBs can exchange the so called Relative Narrowband Transmit Power (RNTP) which is a bitmap containing information on the transmit power on each RB. In the UL there are reactive, using Overload Indicators (OIs) and proactive, using High Interference Indicators (HIIs) methods. For a detailed description see e.g., chapter 12 in [39].

This solution is relatively simple and requires no synchronization of eNBs, only load and scheduling information need to be exchanged. The disadvantage is that the scheduler operating in each eNB can make less optimal scheduling decisions if it has to take neighbor cell interference into account. Moreover, the control channels still interfere, as they are sent on fixed resources. This scheme is slow enough to operate seamlessly on networks with a distributed base station architecture.

In Release 10 eICIC was introduced. eICIC exploits the time domain by introducing Almost Blank Sub-frames (ABSs) meaning that particular sub-frames are muted. (In fact they are not muted completely. To make them backwards compatible with Release 8, some signals, e.g., Common Reference Signal (CRS) is still being transmitted, hence the name *almost* blank). If one transmission is muted, there will be (almost) no interference and this interference-free time interval can now be used to send important information, e.g., signaling and reference signals. The actual muting pattern to use is being coordinated between the eNBs by using the X2 interface. The eICIC concept is standardized, but the actual muting patterns and the algorithms to select them are not.

The power domain can also be exploited to alleviate interference problems. These methods are applicable primarily in the UL direction in HetNet scenarios. The concept is simply to dynamically control the transmit power of the mobile station and in this manner manage interference between the pico and macro layer.

2) *Utilizing Interference Paths Constructively*: The most advanced way of dealing with inter-cell interference is called CoMP, which relies on the fundamental idea to turn interference into a useful signal. This increases the Signal to Interference plus Noise Ratio (SINR) at the mobile, which again turns into higher achievable bit rates. It is included in Release 11 of the specifications [40]–[42].

With CoMP several cells, grouped in a so-called CoMP-set, cooperate on serving one user or a group of users, based on feedback from the mobile(s). Especially in DL this requires tight synchronization and coordination among the base stations in a CoMP set.

The simplest CoMP implementation can be seen as an extension of ICIC. Here one mobile only receives transmission from one eNB (called the serving cell) while the remaining eNBs in the CoMP set aid in avoiding interference. They do that by not using particular sub-carriers (CS—Coordinated Scheduling) and/or utilizing special, e.g., beamforming, antennas (CB—Coordinated Beamforming). Thus, the gain here is that all cells in the CoMP set jointly decide on how to do scheduling and beamforming to minimize interference for all

users. CS/CB requires base station synchronization (0.05 ppm frequency and 3 μ s timing accuracy) similar to ordinary LTE system operation, as only one base station is actively transmitting to one user at a time.

An expansion of CS/CB is called Dynamic Cell Selection (DCS). In this case the data to be transmitted to a particular mobile is made available to all cells in a CoMP set. At a given point of time still only one eNB transmits to a mobile, but the cells coordinate which should do the actual transmission. This is advantageous as transmission can now be done from the eNB which has most favorable transmission path to the mobile. This scheme requires base station synchronization at the same level as CS/CB.

Joint Transmission (JT) [42], [43] is the most advanced CoMP scenario. In JT the data to be transmitted is also available to all cells in the CoMP set, but in this case, several cells jointly and coherently transmit to one user. It relies on very timely and accurate feedback from the terminal on the property of the combined channel from several base stations. To achieve this, a new set of Channel State Information (CSI) reference signal was developed and incorporated into the standards. In single user JT, several cells simply send the same information to one user. Therefore, instead of muting resources (as in ICIC), the same information is transmitted with exact timing to allow the signals to be combined coherently at the receiver and thus achieving a SINR gain. The disadvantage is of course that this takes up resources in several cells and thus effectively creates a reuse factor 1/3 system. This means that it is most suitable for lightly loaded systems. Single user JT can be combined with DCS, meaning that the CoMP set is dynamically changing. For heavily loaded systems JT can be expanded to multiuser JT, where groups of users are sharing (time-frequency) resources. This is, in essence, a combination of multi user MIMO and JT. This scheme requires tight base station synchronization (0.02 ppm frequency and 0.5 μ s timing accuracy) and it is thus beneficial to use in centralized (i.e., C-RAN) based network architectures.

From a performance point of view it turns out that DCS is the best scheme in case of 2×2 MIMO operation. Four transmit antennas are needed to take advantage of more elaborate schemes such as JT.

If all the cells within a CoMP set are served by one BBU Pool, then a single entity doing signal processing enables tighter interaction between base stations. Therefore interference can be kept to lower level and consequently the throughput can be increased [34]. It has been proven that combining clustering of cells with CoMP makes more efficient use of the radio bandwidth [44]. Moreover, ICIC can be implemented over a central unit—BBU Pool—optimizing transmission from many cells to multiple BBUs [43].

In [45], Huiyu *et al.* discusses the factors affecting the performance of CoMP with LTE-A in C-RAN UL, i.e., receiver algorithm, reference signals orthogonality and channel estimation, density and size of the network. In [6], authors present simulation results which compare spectrum efficiency of intra-cell and inter-cell JT to non-cooperative transmission. 13% and 20% increase in spectrum efficiency was observed, respectively. For a cell edge user, spectrum efficiency can increase by 75% and 119%, respectively. In [46], Li *et al.* introduces LTE UL

CoMP joint processing and verify its operation on a C-RAN test bed around Ericsson offices in Beijing. Significant gain was achieved at the cell edge both for intra-site CoMP and inter-site CoMP. Throughput gain is 30–50% when there is no interference and can reach 150% when interference is present. The authors have compared Maximum Ratio Combining (MRC) and full Interference Rejection Combining (IRC). Due to the reduction of X2 usage in C-RAN, real time CoMP can give 10–15% of joint processing gain, while real time ICIC enables 10–30% of multi cell Radio Resource Management (RRM) gain [5]. Performance of multiple-point JT and multiple-user joint scheduling has been analyzed for a non-ideal channel with carrier frequency offset [47]. When carrier frequency offset does not exceed $\pm 3 \sim 5$ ppb, C-RAN can achieve remarkable performance gain on both capacity and coverage even in non-ideal channel, i.e., 20%/52% for cell average/cell edge.

With the introduction of the BBU Pool cooperative techniques, as Multi-Cell MIMO [48] can be enhanced. This can be achieved due to tighter cooperation between base station within a pool. In [49], Liu *et al.* presents a downlink Antenna Selection Optimization scheme for MIMO based on C-RAN that showed advantages over traditional antenna selection schemes.

3) *Decrease of the Delays*: The time needed to perform handovers is reduced as it can be done inside the BBU Pool instead of between eNBs. In [50] Liu *et al.* evaluates the improvement on handover performance in C-RAN and compare it with RAN with RRHs. In Global System for Mobile Communications (GSM), the total average handover interrupt time is lower and the signaling is reduced due to better synchronization of BBUs. In Universal Mobile Telecommunications System (UMTS) signaling, Iub transport bearer setup and transport bandwidth requirements are reduced, however, the performance improvement may not be sensed by the user. For LTE X2-based inter-eNB handover the delay and failure rate are decreased. Moreover, the general amount of signaling information sent to core mobile network is reduced, after being aggregated in the pool.

D. Ease in Network Upgrades and Maintenance

C-RAN architecture with several co-located BBUs eases network maintenance: not only C-RAN capacity peaks and failure might be absorbed by BBU Pool automatic reconfiguration, therefore limiting the need for human intervention, but whenever hardware failures and upgrades are really required, human intervention is to be done only in a very few BBU pool locations. On the contrary for traditional RAN, the servicing may be required at as many cell sites as there are in the network. C-RAN with a virtualized BBU Pool gives a smooth way for introducing new standards, as hardware needs to be placed in few centralized locations. Therefore deploying it can be considered by operators as a part of their migration strategy.

Co-locating BBUs in BBU Pool enables more frequent CPU updates than in case when BBUs are located in remote sites. It is therefore possible to benefit from the IT technology improvements in CPU technology, be it frequency clock (Moore's law) or energy efficiency (as currently seen in Intel mobile processor road map or ARM architecture).

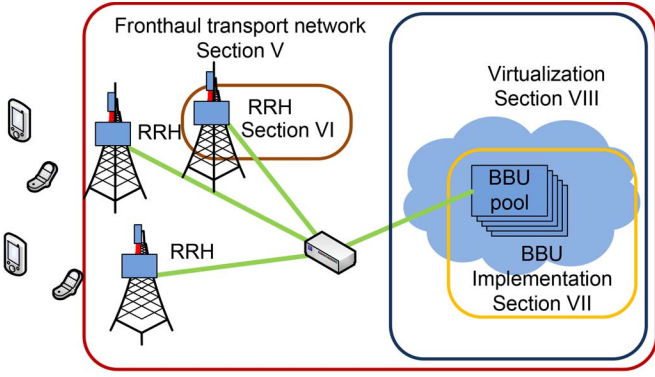


Fig. 8. An overview on technical solutions addressed in this paper.

Software Defined Radio (SDR) is a well-known technology that facilitates implementation in software of such radio functions like modulation/demodulation, signal generation, coding and link-layer protocols. The radio system can be designed to support multiple standards [51]. A possible framework for implementing software base stations that are remotely programmable, upgradable and optimizable is presented in [52]. With such technology, C-RAN BBU Pool can support multi-standard multi-system radio communications configured in software. Upgrades to new frequencies and new standards can be done through software updates rather than hardware upgrades as it is often done today on non-compatible vertical solutions. Multi-mode base station is therefore expected to alleviate the cost of network development and Operations, Administration and Maintenance (OAM).

IV. CHALLENGES OF C-RAN

Before the commercial deployment of C-RAN architectures a number of challenges need to be addressed: A. High bandwidth, strict latency and jitter as well as low cost transport network needs to be available, B. Techniques on BBU cooperation, interconnection and clustering need to be developed as well as C. Virtualization techniques for BBU Pool need to be proposed. In this section we elaborate on those challenges. In the latter sections we present an ongoing work on possible technical solutions that enable C-RAN implementation (Sections V–VIII). Fig. 8 gives an overview of technical solutions addressed in the article.

A. Need for High Bandwidth, Strict Latency and Jitter as Well as Low Cost Transport Network

C-RAN architecture brings a huge overhead on the optical links between RRH and BBU Pool. Comparing with backhaul requirements, the one on fronthaul are envisioned to be 50 times higher [43].

IQ data is sent between BBU and RRH as shown in Fig. 3. The main contributors to the size of IQ data are: turbocoding (e.g., in UMTS and LTE 1:3 turbocode is used resulting in three times overhead), chosen radio interface (e.g., CPRI) IQ sample width and oversampling of LTE signal. For example, 30.72 MHz sampling frequency is standardized for 20 MHz LTE, which is more than 20 MHz needed according to Nyquist–Shannon sampling theorem. Total bandwidth

TABLE II
IQ BIT RATES BETWEEN A CELL SITE AND CENTRALIZED BBU POOL

Cell configuration	Bit rate	Source
20 MHz LTE, 15+1 CPRI IQ Sample width, 10/8 line coding, 2x2 MIMO	2.5 Gbps	
5x20 MHz LTE-A, 15 CPRI IQ Sample width, 2x2 MIMO, 3 sectors	13.8 Gbps	[53]
20 MHz LTE, 4x2 MIMO, 3 sectors	16.6 Gbps	[10]
TD-LTE, 3 sectors	30 Gbps	[54]
1.6 MHz TD-SCDMA, 8Tx/8Rx antennas, 4 times sampling rate	330 Mbps	[6]
TD-SCDMA S444, 3 sectors	6 Gbps	[54]
200 kHz GSM, 2Tx/2Rx antennas, 4x sampling rate	25.6 Mbps	[6]

depends also on number of sectors and MIMO configuration. Equation (1) summarizes factors that influence IQ bandwidth. Scenario of 20 MHz LTE, 15 + 1 CPRI IQ Sample width, 10/8 line coding, 2×2 MIMO transmission resulting in 2.5 Gbps bit rate in fronthal link is often treated as a baseline scenario. Consequently, for 20 MHz 4×4 MIMO, 3 sector base station, the expected IQ throughput exceeds 10 Gbps. Examples on expected IQ bit rate between cell site and BBU in LTE-A, LTE, Time Division Synchronous Code Division Multiple Access (TD-SCDMA) and GSM networks can be found in Table II. The centralized BBU Pool should support 10–1000 base station sites [6], therefore a vast amount of data needs to be carried towards it.

$$IQBandwidth = samplingFrequency \cdot sampleWidth$$

$$\cdot 2 \cdot lineCoding \cdot MIMO \cdot noOfSectors. \quad (1)$$

The transport network not only needs to support high bandwidth and be cost efficient, but also needs to support strict latency and jitter requirements. Below different constraints on delay and jitter are summarized:

- 1) The most advanced CoMP scheme, JT, introduced in Section III-C requires $0.5 \mu s$ timing accuracy in collaboration between base stations, which is the tightest constraint. However, it is easier to cope with synchronization challenges in C-RAN compared to traditional RAN due to the fact that BBUs are co-located in the BBU Pool.
- 2) According to [6], regardless of the delay caused by the cable length, round trip delay of user data may not exceed $5 \mu s$, measured with the accuracy of ± 16.276 ns on each link or hop [20].
- 3) The sub-frame processing delay on a link between RRHs and BBU should be kept below 1 ms, to meet HARQ requirements. Due to the delay requirements of HARQ mechanism, generally maximum distance between RRH and BBU must not exceed 20–40 km [6].

Recommendations on transport network capacity can be found in Section V.

B. BBU Cooperation, Interconnection and Clustering

Cooperation between base stations is needed to support CoMP in terms of sharing the user data, scheduling at the base station and handling channel feedback information to deal with interference.